Nios V ↔ Caliptra 2.x Integration Guide (passive / core-only)

> **Where the files are (published copy).** This guide was written in the local Caliptra 2.x design tree. In this repo, the 2.x sources it names live under `caliptra2x/`:
>
> - `rtl_2x/`, `sim_2x/`, `sim/rst_release_stub.sv`, `fw_2x/`, `scripts/gen_fuse_image_2x.py` and `quartus_a5e_2x/` → `caliptra2x/…`
> - `docs/P1_FIT_RESULT_2X_A5E.md` and `docs/CALIPTRA_2X_INTEGRATION_PLAN.md` → this folder
> - The 1.x guide `docs/NIOS_CALIPTRA_INTEGRATION.md` is `../NIOS_CALIPTRA_INTEGRATION.md`
>
> caliptra-rtl and caliptra-sw are not vendored.

- **Project:** Caliptra 2.x in passive (core-only) mode on Intel Agilex 5 E **A5EC065BB32AE6S**, with a Nios V/m mailbox host (Avalon → AXI4).
- **RTL:** chipsalliance/caliptra-rtl **v2.1.2**, commit `49370266d12cb0c4a8f71b3a0ff7e54ba7d4866e`. `CALIPTRA_MODE_SUBSYSTEM` is **not** defined (core-only), and `CALIPTRA_INTERNAL_TRNG` is **not** defined (external TRNG).
- **Firmware:** caliptra-sw `dd1e79d5bf55a9bd5bdb05915757eaf34844145d`, which carries tags `rom-2.1.3-1` and `fw-2.1.4`. See FIRMWARE_AND_KEYS_2X.md.
- **Simulation:** Verilator 5.054, one cold boot to `ready_for_runtime`, **PASS**. See SIMULATION_2X.md.
- **Companion docs:** docs/P1_FIT_RESULT_2X_A5E.md (Quartus fit gate), docs/CALIPTRA_2X_INTEGRATION_PLAN.md (1.x → 2.x delta and phased plan). The 1.x guide this one is modelled on is `docs/NIOS_CALIPTRA_INTEGRATION.md` (rev 3).
- **Date:** 2026-11-10 (America/Chicago).

> **Status.** This is a *design study*. The 2.x glue RTL and the boot flow are verified in Verilator. The design **does not fit** the A5E 065: synthesis estimates 279,685 / 222,400 ALMs (126 %) (§14). Every key, seed and entropy value is a **TEST/STUDY value**: the obfuscation key, CSR HMAC key, UDS, field entropy, the test-signed firmware, and the simulated TRNG data. The firmware is signed with caliptra-sw's **public fake keys**. See FIRMWARE_AND_KEYS_2X.md §6.

What exists and what doesn't (2.x tree):

Item	State
`rtl_2x/nios_caliptra2x_top.sv`, `avmm_to_axi.sv`, `soc_fuse_ctrl2x.sv`, `soc_fw_store.sv`, `reset_sync.sv`, `caliptra_passive_top.sv`	Written; simulated (Verilator)
`scripts/gen_fuse_image_2x.py` → `fw_2x/fuse_image_2x.hex`	Written; output used in the PASS run (regeneration is byte-identical)
`fw_2x/` ROM + FW bundle (+ `build_rom.sh`, `SHA256SUMS`)	Built from caliptra-sw `dd1e79d`
Platform Designer system `nios_caliptra2x_sys` (Nios V + PIOs + 1 MiB window)	**Not created yet.** The TB uses a behavioral stand-in with the same ports
Nios C firmware for 2.x (`main.c` port)	**Not ported yet.** The 1.x `quartus_a5e_nios/sw/app/main.c` is unchanged. The TB host model defines the 2.x sequence (§11)
Quartus project for the full Nios + 2.x top	**Not created.** `quartus_a5e_2x/` synthesizes only the Caliptra core wrapper (fit gate)

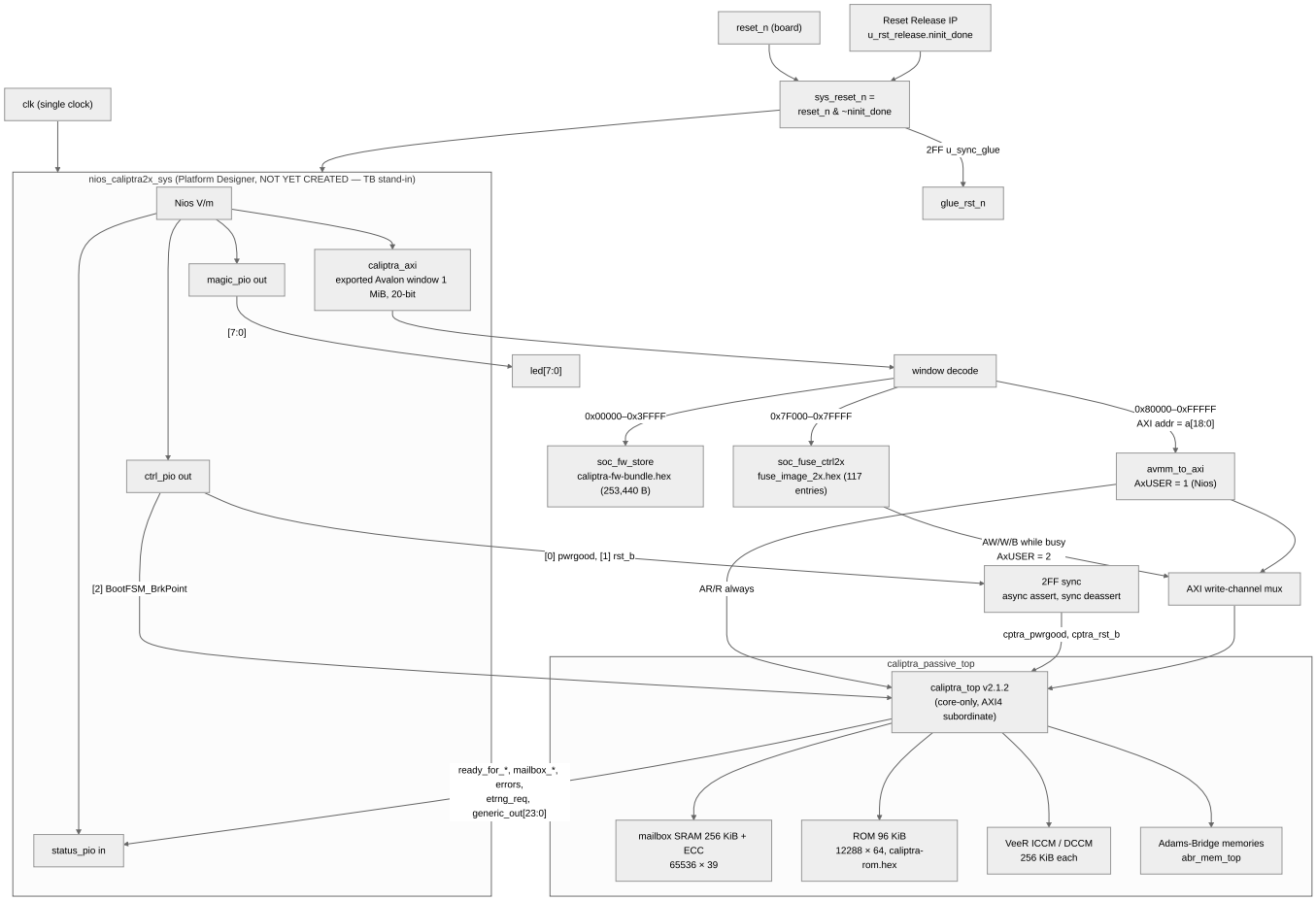
---

1. Source files

File	Role
`rtl_2x/nios_caliptra2x_top.sv`	FPGA top: Reset Release gate, PD system, SoC window decode, Avalon→AXI bridge, fuse controller, AXI write-channel mux, 2FF reset syncs, straps, status PIO
`rtl_2x/caliptra_passive_top.sv`	Core-only wrapper around `caliptra_top`: flattens the AXI subordinate to ports, ties off the AXI manager (DMA) and all `ss_*` straps, instantiates the mailbox SRAM, ROM, VeeR ICCM/DCCM and Adams-Bridge memories
`rtl_2x/avmm_to_axi.sv`	Avalon-MM agent → AXI4 manager, single beat, 1 outstanding, AxUSER = `user_i`
`rtl_2x/soc_fuse_ctrl2x.sv`	SoC fuse sequencer (stand-in for OTP + fuse controller); replays `fw_2x/fuse_image_2x.hex` over AXI with AxUSER 2
`rtl_2x/soc_fw_store.sv`	Read-only M20K store holding `fw_2x/caliptra-fw-bundle.hex` (stand-in for boot flash)
`rtl_2x/reset_sync.sv`	`rst_sync_async_assert` / `pwrgood_sync_async_assert` (async assert, 2FF sync deassert)
`rtl_2x/caliptra_veer_sram_export.sv`	VeeR ICCM/DCCM SRAM export
`sim/rst_release_stub.sv`	Verilator stand-in for the Agilex 5 Reset Release IP (`ninit_done` high for 25 ns)
`scripts/gen_fuse_image_2x.py`	Builds the fuse image from the FW bundle
`fw_2x/`	ROM, FW bundle, hex files, `SHA256SUMS`, `build_rom.sh`, `fuse_image_2x.hex`

File	Role
`sim_2x/`	`boot2x.f`, `tb_nios2x_top_fw.sv`, `run_nios2x_top_fw.sh`, `BOOT_NIOS2X.md`, `nios2x_top_fw_run.log`

2. Block diagram



3. Port and signal tables

3.1 `nios_caliptra2x_top` parameters and ports

Parameter	Default	Meaning
`ROM_INIT_FILE`	`"../fw_2x/caliptra-rom.hex"`	Caliptra ROM image (`$readmemh`, 64-bit words)
`FW_INIT_FILE`	`"../fw_2x/caliptra-fw-bundle.hex"`	FW bundle for `soc_fw_store` (32-bit LE words)
`FUSE_IMAGE`	`"../fw_2x/fuse_image_2x.hex"`	Fuse image for `soc_fuse_ctrl2x`
`FW_BYTES`	`253440`	Bundle size reported by fuse ctrl MMIO `0x10` (used as `MBOX_DLEN`)
`NUM_FUSES`	`117`	Image entries (104 fuse words + 12 owner-hash words + 1 lock)
`NIOS_AXI_USER`	`32'h1`	AxUSER driven on every Nios access
`FUSE_AXI_USER`	`32'h2`	AxUSER driven by the fuse controller

Port	Dir	Width	Connection
`clk`	in	1	Single clock for Nios, glue and Caliptra
`reset_n`	in	1	Board reset, active low; ANDed with `~ninit_done`
`led`	out	8	`magic_pio[7:0]`

3.2 Internal glue signals

Signal	Source → sink	Notes
`ninit_done`	`rst_release` → gate	Agilex 5 Reset Release IP (sim: `sim/rst_release_stub.sv`)
`sys_reset_n`	`reset_n & ~ninit_done`	PD reset; also gates the raw pwrgood/rst_b requests
`glue_rst_n`	`rst_sync_async_assert` (2 stages)	Resets decode, bridge, fuse ctrl
`ctrl_out[0]`	ctrl PIO → `raw_pwrgood = ctrl_out[0] & sys_reset_n` → `pwrgood_sync_async_assert` → `cptra_pwrgood`	Reset value 0 = Caliptra unpowered
`ctrl_out[1]`	ctrl PIO → `raw_rst_b = ctrl_out[1] & sys_reset_n` → `rst_sync_async_assert` → `cptra_rst_b`	Warm reset = drop only this bit
`ctrl_out[2]`	ctrl PIO → `BootFSM_BrkPoint`	Default 0
`status_in`	→ status PIO	See §3.3
`fuse_busy`	`soc_fuse_ctrl2x.busy`	1 in NEXT/SEND/RESP: fuse ctrl owns AW/W/B
`bridge_idle`	`avmm_to_axi.idle`	Exported for a warm-reset quiesce; **not yet used** in the top
`nios_axi_user`	register, reset/load `NIOS_AXI_USER`	A register (not an assign) so the TB can `force` a wrong ID
`generic_input_wires`	`64'b0`	IS Table 13: bit 0 = zeroization status; no source → 0

3.3 `status_pio` bits (same positions as 1.x)

Bit	Signal	1.x name
0	`ready_for_fuses`	same
1	`ready_for_mb_processing`	`ready_for_fw_push`
2	`ready_for_runtime`	same
3	`mailbox_data_avail`	same
4	`mailbox_flow_done`	same
5	`cptra_error_fatal`	same
6	`cptra_error_non_fatal`	same
7	`etrng_req`	same
31:8	`generic_output_wires[23:0]`	same (ROM/FW log byte on `[7:0]`, strobe toggle on `[8]`)

3.4 `caliptra_passive_top` / `caliptra_top` connections

Widths come from `config_defines.svh`: `CALIPTRA_AXI_DATA_WIDTH` = 32, `CALIPTRA_AXI_USER_WIDTH` = 32, `CALIPTRA_AXI_ID_WIDTH` = 8. The SoC-interface address width is `CALIPTRA_SLAVE_ADDR_WIDTH(SOC_IFC)` = **19** bits (base `0x3000_0000`, mask `0x3007_FFFF`).

Clocks, resets, straps, keys

`caliptra_top` port	Width	Driven by (`nios_caliptra2x_top` → wrapper)	Status
`clk`	1	`clk`	
`cptra_pwrgood`	1	2FF-synced `ctrl_out[0]`	
`cptra_rst_b`	1	2FF-synced `ctrl_out[1]`	
`cptra_obf_key`	256	`STUDY_PROD_MIMIC_OBF_KEY` = `31358e8a_f34d6a45_…_b2c3b8d5` (localparam)	**TEST-ONLY**
`cptra_csr_hmac_key`	16 × 32 (`CLP_CSR_HMAC_KEY_DWORDS` = 16)	`{16{32'h5A3C_96E1}}`	**TEST-ONLY**
`security_state.device_lifecycle`	2	`2'b11` = DEVICE_PRODUCTION	**TEST-ONLY** (hardwired)
`security_state.debug_locked`	1	`1'b1`	**TEST-ONLY** (hardwired)
`cptra_obf_field_entropy_vld` / `cptra_obf_field_entropy`	1 / 256	`0` / `'0`	Passive: tie 0 (IS Table 12)
`cptra_obf_uds_seed_vld` / `cptra_obf_uds_seed`	1 / 512	`0` / `'0`	Passive: tie 0
`BootFSM_BrkPoint`	1	`ctrl_out[2]`	
`scan_mode`	1	`1'b0`	

caliptra_top port	Width	Driven by (nios_caliptra2x_top → wrapper)	Status
jtag_tck/tms/tdi/trst_n	1 each	0 (TAP held in reset)	OPEN (no TAP endpoint)
jtag_tdo, jtag_tdoEn	1 each	unconnected	

AXI4 subordinate (s_axi_w_if, s_axi_r_if), flattened in the wrapper

Channel	Signals	Driven by
AW	awaddr[18:0], awburst, awsize, awlen, awuser[31:0], awid[7:0], awlock, awvalid / awready	Mux: fuse ctrl while fuse_busy, else bridge. awburst/awsize/awlen/awid/awlock always come from the bridge (constants INCR / 2 / 0 / 0 / 0)
W	wdata[31:0], wstrb, wuser, wvalid / wready, wlast	Mux for wdata/wvalid/wuser; wstrb=4'hF, wlast=1 from bridge
B	bresp, bid, buser, bvalid / bready	bresp/bvalid to both; bready muxed; bid/buser unconnected
AR	araddr, arburst, arsize, arlen, aruser, arid, arlock, arvalid / arready	Bridge only
R	rdata, rresp, rid, ruser, rlast, rvalid / rready	Bridge only; rid/ruser/rlast unconnected

AXI manager (DMA), subsystem straps and other ports, all tied off in the wrapper

Port group	Tie-off	Reason
m_axi_w_if, m_axi_r_if	awready/wready/arready/bvalid/rvalid = 0, data/resp/id/user = 0	Passive: DMA AXI manager must not reach the SoC
strap_ss_caliptra_base_addr, strap_ss_mci_base_addr, strap_ss_recovery_ifc_base_addr, strap_ss_external_staging_area_base_addr, strap_ss_otp_fc_base_addr, strap_ss_uds_seed_base_addr, strap_ss_key_release_base_addr, strap_ss_key_release_key_size, strap_ss_prod_debug_unlock_auth_pk_hash_reg_bank_offset, strap_ss_num_of_prod_debug_unlock_auth_pk_hashes, strap_ss_caliptra_dma_axi_user, strap_ss_strap_generic_0..3	'0	IS Table 11: subsystem only; tie 0 in passive
ss_debug_intent	1'b0	Subsystem only
ss_ocp_lock_en	1'b0 (constant)	OCP LOCK needs subsystem; the runtime log confirms [rt] OCP LOCK Unsupported
recovery_data_avail, recovery_image_activated	1'b0	Subsystem recovery interface
ss_dbg_manuf_enable, ss_soc_dbg_unlock_level, ss_generic_fw_exec_ctrl	unconnected outputs	
trace_rv_*	unconnected	
itrng_data[3:0], itrng_valid	4'h0, 1'b0	Internal TRNG not compiled in
generic_input_wires[63:0]	64'b0	
generic_output_wires[63:0]	[23:0] → status_pio[31:8]	Log channel

Status and interrupts

Port	Width	To
ready_for_fuses, ready_for_mb_processing, ready_for_runtime	1 each	status_pio[0..2]; ready_for_fuses also to the fuse ctrl
mailbox_data_avail, mailbox_flow_done	1 each	status_pio[3..4] (polled, no IRQ)
cptra_error_fatal, cptra_error_non_fatal	1 each	status_pio[5..6] (polled, no IRQ)
etrng_req	1	status_pio[7]

Memories (instantiated inside caliptra_passive_top)

Memory	Shape	Module
Mailbox SRAM	CPTRA_MBOX_DEPTH × CPTRA_MBOX_DATA_AND_ECC_W = 65536 × 39 (256 KiB data + 7-bit ECC; passive size)	caliptra_sram

Memory	Shape	Module
ROM (imem)	`CALIPTRA_IMEM_DEPTH` × 64 = 12288 × 64 (96 KiB, `CALIPTRA_IMEM_BYTE_SIZE` = 98304)	`caliptra_rom_m20k`, `$readmemh(ROM_INIT_FILE)`
VeeR ICCM / DCCM	256 KiB each	`caliptra_veer_sram_export`
Adams-Bridge (ML-DSA / ML-KEM)	10 memories via `abr_mem_if`	`abr_mem_top`

---

4. AXI user IDs (AxUSER)

Agent	AxUSER	Where set	Programmed into
Nios (bridge)	`0x0000_0001`	`NIOS_AXI_USER` → `nios_axi_user` → `avmm_to_axi.user_i` (AWUSER, WUSER, ARUSER)	`CPTRA_MBOX_VALID_AXI_USER[0]` + lock; `CPTRA_TRNG_VALID_AXI_USER` + lock
Fuse controller	`0x0000_0002`	`FUSE_AXI_USER` → `soc_fuse_ctrl2x.axi_user` (AWUSER, WUSER)	`CPTRA_FUSE_VALID_AXI_USER` + lock
(reserved)	`0xFFFF_FFFF`	never driven	IS Table 19: "CANNOT use value 0xFFFFFFFF". It is also the RTL default `CPTRA_DEF_MBOX_VALID_AXI_USER`

- The IDs are unique per agent and non-zero, as IS Table 19 recommends.
- The RTL integration parameters are left at upstream defaults: `CPTRA_SET_MBOX_AXI_USER_INTEG = 0`, `CPTRA_SET_FUSE_AXI_USER_INTEG = 0`. The IDs are therefore *programmed and locked* by the host on every cold boot (§11, step 4), not hardened at compile time.
- **Fuse range gating.** In `soc_ifc_arb.sv`, SoC requests to `0x30200–0x305FF` (`SOC_IFC_FUSE_START_ADDR`..`_END_ADDR`) are routed only if `valid_fuse_user` is set. Once `CPTRA_FUSE_AXI_USER_LOCK` = 1, that means AxUSER == `CPTRA_FUSE_VALID_AXI_USER` (= 2). So **the Nios (user 1) can neither write nor read fuse registers**. This explains the `FUSE_PQC_KEY_TYPE readback=00000000` seen in the sim (§9.3).
- `CPTRA_OWNER_PK_HASH[12]` (`0x30140`) and `CPTRA_OWNER_PK_HASH_LOCK` (`0x30170`) are **outside** the fuse range. Any SoC requester can write them until the lock is set. In this design the fuse controller writes and locks them. Note that the Nios could also write them before the lock (see checklist, §13).
- The sim's `+test=axiuser` checks: a mailbox write with AxUSER 3 gives SLVERR; a fuse write with AxUSER 3 gives SLVERR; a fuse write with AxUSER 1 (Nios) gives SLVERR; a `HW_REV_ID` read with AxUSER 1 is accepted (`TB_AXIUSER PASS`).

---

5. Memory map

5.1 Nios SoC window (20-bit, 1 MiB) — decode in `nios_caliptra2x_top`

The window base in the Nios address space is set by the (not yet created) Platform Designer system. Offsets below are **window offsets** (`cl_address[19:0]`).

Window offset	Decode (RTL)	Target
`0x00000–0x3FFFF`	`!a[19] && !a[18]`	`soc_fw_store`: 65536 × 32-bit words (256 KiB), read-only, 1-cycle read. The 2.x bundle uses 63,360 words (253,440 B)
`0x40000–0x7EFFF`	—	Unmapped: no waitrequest; read data is not meaningful
`0x7F000–0x7FFFF`	`!a[19] && a[18:12]==7'h7F`	`soc_fuse_ctrl2x` MMIO (§7.2)
`0x80000–0xFFFFF`	`a[19]`	Caliptra AXI subordinate; **AXI address = `a[18:0]`**

> The RTL comment in `nios_caliptra2x_top.sv` says "0x80000-0xBFFFF … AXI addr = cl_address[17:0]". The actual decode is `sel_cptra = cl_address[19]`, and the bridge takes `cl_address[AXI_AW-1:0]` with `AXI_AW` = 19. So the whole upper half maps through. Caliptra's targets all sit below AXI `0x40000` for the SoC (see 5.2), so the narrower comment is harmless, but this table follows the RTL.

5.2 Caliptra 2.x SoC AXI map (from `caliptra_reg_defines.svh` / `soc_ifc_pkg.sv`, v2.1.2)

Internal addresses are `0x3000_0000 + AXI address` (`SOC_IFC_REG_OFFSET`).

AXI address	Nios window offset	Block	SoC access
`0x20000–0x20FFF`	`0xA0000`	`mbox_csr`	Valid mailbox AxUSERs
`0x21000–0x21FFF`	`0xA1000`	SHA512 accelerator CSR	Only the `SS_CALIPTRA_DMA_AXI_USER` → not usable by the Nios in passive mode
`0x22000–0x22FFF`	`0xA2000`	AXI DMA registers	Present in RTL; DMA unused in passive mode

AXI address	Nios window offset	Block	SoC access
0x30000–0x3FFFF	0xB0000	soc_ifc registers	Any requester except the fuse range
0x30200–0x305FF	0xB0200	Fuse registers (subset of soc_ifc)	Only CPTRA_FUSE_VALID_AXI_USER once locked

5.3 Registers used by the host (verified against caliptra_reg_defines.svh)

Register	AXI	Window	Use
MBOX_LOCK	0x20000	0xA0000	Read to acquire (bit 0 = 0 → acquired)
MBOX_USER	0x20004	0xA0004	Expect 0x1
MBOX_CMD	0x20008	0xA0008	0x46574C44 = FIRMWARE_LOAD
MBOX_DLEN	0x2000C	0xA000C	253,440
MBOX_DATAIN	0x20010	0xA0010	Full-word writes only
MBOX_DATAOUT	0x20014	0xA0014	(not used for FW load)
MBOX_EXECUTE	0x20018	0xA0018	1 = execute, 0 = release
MBOX_STATUS	0x2001C	0xA001C	[3:0] 2 = CMD_COMPLETE
CPTRA_HW_ERROR_FATAL / _NON_FATAL	0x30000 / 0x30004	0xB0000 / 0xB0004	RW1C
CPTRA_FW_ERROR_FATAL / _NON_FATAL	0x30008 / 0x3000C	0xB0008 / 0xB000C	RW1C
CPTRA_BOOT_STATUS	0x30038	0xB0038	
CPTRA_FLOW_STATUS	0x3003C	0xB003C	
CPTRA_SECURITY_STATE	0x30044	0xB0044	Sim read 0x00000007 = Production + debug locked
CPTRA_MBOX_VALID_AXI_USER[0] / CPTRA_MBOX_AXI_USER_LOCK[0]	0x30048 / 0x3005C	0xB0048 / 0xB005C	1 / 1
CPTRA_TRNG_VALID_AXI_USER / CPTRA_TRNG_AXI_USER_LOCK	0x30070 / 0x30074	0xB0070 / 0xB0074	1 / 1
CPTRA_TRNG_DATA[0..11]	0x30078–0x300A4	0xB0078–0xB00A4	12 entropy words
CPTRA_TRNG_CTRL	0x300A8	0xB00A8	(not used)
CPTRA_TRNG_STATUS	0x300AC	0xB00AC	Write 2 = DATA_WR_DONE
CPTRA_FUSE_WR_DONE	0x300B0	0xB00B0	Written 1 by the fuse ctrl (its FUSE_WR_DONE parameter 32'h0003_00B0)
CPTRA_TIMER_CONFIG	0x300B4	0xB00B4	20000 (ps period for 50 MHz)
CPTRA_BOOTFSM_GO	0x300B8	0xB00B8	(only with BootFSM_BrkPoint =1)
CPTRA_HW_REV_ID	0x300D4	0xB00D4	Sim read 0x00000212
CPTRA_HW_CONFIG	0x300E0	0xB00E0	Sim read 0x00000010 (iTRNG_en =0, SUBSYSTEM_MODE_en =0)
CPTRA_FUSE_VALID_AXI_USER / CPTRA_FUSE_AXI_USER_LOCK	0x30108 / 0x3010C	0xB0108 / 0xB010C	2 / 1
CPTRA_WDT_CFG[0] / [1]	0x30110 / 0x30114	0xB0110 / 0xB0114	100,000,000 / 0
CPTRA_OWNER_PK_HASH[0..11]	0x30140–0x3016C	—	Written by the fuse ctrl
CPTRA_OWNER_PK_HASH_LOCK	0x30170	0xB0170	Written 1 by the fuse ctrl

6. avmm_to_axi bridge

- **Avalon side:** waitrequest + readdatavalid agent, with no bursts and no pipelining. avs_byteenable is ignored because Caliptra SoC registers are 32-bit.
- **AXI side:** single beat, AxLEN=0, AxSIZE=2 (32-bit), AxBURST=INCR (2'b01), AxLOCK=0, AxID=0, WSTRB=4'hF, WLAST=1. AWUSER = WUSER = ARUSER = user_i. The address is word-aligned ({a[AW-1:2],2'b00}).
- **One outstanding transaction.** States: IDLE → WADDR → WRESP → DONE for writes and IDLE → RADDR → RDATA → DONE → RESP for reads. AW and W are issued together, and each drops on its own handshake.
- **Avalon handshake:** avs_waitrequest is low for exactly one cycle (DONE) to accept. Read data comes one cycle later with avs_readdatavalid (RESP).

- **Errors:** `BRESP`/`RRESP` ≠ OKAY sets `err_q`, so `avs_response = 2'b10` (SLVERR) for the last transaction. The TB reads `u_bridge.err_q` directly for the AxUSER test. Firmware would read the Avalon response, if the PD interconnect forwards it.
- **Spec fit (IS "Unsupported features"):** single outstanding, in order, no narrow writes to `mbox_datain` / `TRNG_DATA`. AxCACHE/PROT/REGION/QOS are not driven.
- **`idle` output:** high in `IDLE`, intended for warm-reset quiescing. It is not yet used by the top.

---

7. `soc_fuse_ctrl2x` — fuse sequencer

7.1 Image format (`fw_2x/fuse_image_2x.hex`)

- Each line is one `$readmemh` 64-bit word, `{AXI byte address[31:0], data[31:0]}`, for example the first line `00030200811944f5` (`FUSE_UDS_SEED[0]`) and the last line `0003017000000001` (`CPTRA_OWNER_PK_HASH_LOCK` = 1). Addresses are explicit because the 2.x fuse map has holes.
- The controller truncates the address to `AW` = 19 bits.
- 117 entries: 104 fuse words, then 12 `CPTRA_OWNER_PK_HASH` words, then `CPTRA_OWNER_PK_HASH_LOCK = 1`. The controller then appends `CPTRA_FUSE_WR_DONE = 1` itself (`addr = FUSE_WR_DONE = 0x300B0`). That makes **118 AXI writes**.

7.2 MMIO (window `0x7F000`, 4 KiB, word registers)

Offset	Name	Access	Bits
`0x00`	ID	RO	`0x4655_5345` ("FUSE")
`0x04`	CTRL	WO	bit0 **START**, bit1 **CLEAR** (clears DONE/ERR), bit2 **START_DONE**
`0x08`	STATUS	RO	bit0 BUSY, bit1 DONE, bit2 ERR, bit3 READY_FOR_FUSES
`0x0C`	COUNT	RO	AXI writes completed (118 after a full START)
`0x10`	FW_BYTES	RO	`FW_BYTES` parameter (253,440)
`0x14`	NUM_FUSES	RO	`NUM_FUSES` parameter (117)

7.3 Behaviour

- **START** (cold boot) replays entries 0..116, then `FUSE_WR_DONE`.
- **START_DONE** (warm reset) starts with `idx = NUM_FUSES`, which is "last", so it writes **only** `FUSE_WR_DONE = 1`. Fuse registers and `CPTRA_OWNER_PK_HASH*` are pwrgood-domain and persist across a warm reset. START wins if both bits are set.
- START or START_DONE while `ready_for_fuses` = 0 leads to `ERR`, with no writes.
- Any write with `BRESP` ≠ OKAY leads to `ERR`, and the sequence stops (`COUNT` shows how far it got).
- While `busy` is set (NEXT/SEND/RESP), the top muxes AW/W/B to the fuse controller. The Nios only polls the fuse ctrl STATUS (a local slave) meanwhile, so the bridge is idle. The read channels always belong to the bridge.
- Measured in sim: START issued just after cycle 136, `STATUS=00000002 COUNT=118` at cycle 614, so about 4 cycles per write.

---

8. Fuse map decode (what `gen_fuse_image_2x.py` writes)

Offsets are verified against `caliptra_reg_defines.svh` (`CLP_SOC_IFC_REG_*`, minus `0x3000_0000`). The access column is from `soc_ifc_fuse_reg.rdl` / `soc_ifc_external_reg.rdl`.

AXI addr	Register	Words	Value written	Notes
`0x30200`	`FUSE_UDS_SEED[0..15]`	16	STUDY: `0x1F2E3D4C ^ (0x9E3779B9·(i+1))` → `811944f5 2340ce3e … fc59a6dc`	**512-bit** in 2.x (384 in 1.x). Caliptra: no access; SoC: WL-S (write-only). **TEST-ONLY**
`0x30240`	`FUSE_FIELD_ENTROPY[0..7]`	8	STUDY: `0xA7C3E519 ^ (0x85EBCA6B·(i+1))` → `22282f72 ac1471cf … 889db641`	256-bit. **TEST-ONLY**
`0x30260`	`FUSE_VENDOR_PK_HASH[0..11]`	12	SHA-384(`preamble.vendor_pub_key_info`) as 12 big-endian words → `30399676 a17e3e97 3677b3ff 862f4bf2 d1932d88 4778453c 376fe00d c93fb8aa 0770f3eb f3411a08 53e9c57e ce8a2980`	Hash of the ECC + PQC key *descriptors* (1,736 B). ROM rejects an all-zero value
`0x30290`	`FUSE_ECC_REVOCATION`	1	0	4-bit field
`0x302B4`	`FUSE_FMC_KEY_MANIFEST_SVN`	1	0	

AXI addr	Register	Words	Value written	Notes
`0x302B8`	`FUSE_RUNTIME_SVN[0..3]`	4	0	Bundle header SVN must be ≥ fuse SVN; the bundle SVN is 0
`0x302C8`	`FUSE_ANTI_ROLLBACK_DISABLE`	1	0	
`0x302CC`	`FUSE_IDEVID_CERT_ATTR[0..23]`	24	0	
`0x3032C`	`FUSE_IDEVID_MANUF_HSM_ID[0..3]`	4	0	
`0x30340`	`FUSE_LMS_REVOCATION`	1	0	
`0x30344`	`FUSE_MLDSA_REVOCATION`	1	0	
`0x30348`	`FUSE_SOC_STEPPING_ID`	1	1	Test value
`0x3034C`	`FUSE_MANUF_DBG_UNLOCK_TOKEN[0..15]`	16	0	Debug unlock is subsystem-only
`0x3038C`	`FUSE_PQC_KEY_TYPE`	1	**1** = manifest `pqc_key_type`	RDL: 2-bit, "one-hot", bit0 = MLDSA, bit1 = LMS. The ROM compares it with `FwVerificationPqcKeyType::from_u8()` (MLDSA = 1, LMS = 3). Skipped only when Unprovisioned
`0x30390`	`FUSE_SOC_MANIFEST_SVN[0..3]`	4	0	Unused in passive
`0x303A0`	`FUSE_SOC_MANIFEST_MAX_SVN`	1	0	
`0x303C0`	`FUSE_HEK_SEED[0..7]`	8	0	OCP LOCK; unused (`ss_ocp_lock_en`=0)
	Fuse words subtotal	**104**		Holes at `0x294–0x2B0` and `0x3A4–0x3BC` are skipped
`0x30140`	`CPTRA_OWNER_PK_HASH[0..11]`	12	SHA-384(`preamble.owner_pub_keys`) → `48afdb07 3c5e0d4e e4649046 8ef81f2c f57249b6 e76a28f5 fca4de69 6a7d3e2e d3efc4e6 77431854 3e95307a 54988bd7`	**Not a fuse in 2.x.** Architectural register, `resetsignal = cptra_pwrgood`, SoC RWL-S
`0x30170`	`CPTRA_OWNER_PK_HASH_LOCK`	1	1	RW1-S; reset only by power cycle
`0x300B0`	`CPTRA_FUSE_WR_DONE`	(1)	1	Appended by the controller, not in the image

Owner key and entropy, in plain words:

- The **owner key hash is provisioned**, written to `CPTRA_OWNER_PK_HASH` and locked. The sim read back `[0]=48afdb07` and `LOCK=1`. The ROM then verified the image against it.
- **UDS and field entropy are provisioned** with STUDY values (not secrets). The SoC cannot read them back (write-only), and the Nios can't access the fuse range anyway.
- There is no `FUSE_LIFE_CYCLE` in 2.x. Lifecycle comes only from the `security_state` straps (Production, debug locked).

---

9. TRNG (external, serviced by the host)

9.1 Hardware path

- `CALIPTRA_INTERNAL_TRNG` is not defined. `CPTRA_HW_CONFIG.iTRNG_en` = 0. `itrng_*` are tied 0.
- When Caliptra firmware wants entropy, it raises `etrng_req` (`status_pio[7]`).
- The host writes 12 words to `CPTRA_TRNG_DATA[0..11]` (`0x30078–0x300A4`), then writes `CPTRA_TRNG_STATUS = 2` (`DATA_WR_DONE`, `0x300AC`).
- Writes must come from `CPTRA_TRNG_VALID_AXI_USER` (= 1, locked). `DATA_WR_DONE` is hardware-cleared when Caliptra FW clears `DATA_REQ` (`soc_ifc_top.sv`).

9.2 Entropy source — TEST-ONLY

- **Sim:** the TB host model writes `$urandom` values. In the 1.x Nios firmware, `main.c` uses `xorshift32` seeded `0x1234ABCD`. The 2.x Nios firmware port would keep that (it isn't ported yet).
- Neither source is a real entropy source. IS §TRNG: *"It is a requirement for FIPS compliance that this TRNG HW API is always handled by SoC hardware gasket logic (and not some SoC ROM or firmware code)."*
- **Production options:** a hardware ETRNG gasket fed by a physical noise source, or `CALIPTRA_INTERNAL_TRNG` with a PTRNG on `itrng_data/itrng_valid`.

9.3 Sim numbers

- **217** requests serviced in one boot.
- The first was at cycle 379,157.

- 21 had been serviced by `ready_for_mb_processing`.
- The TB fails if `etrng_req` stays high for 100,000 cycles after `DATA_WR_DONE`; that never happened.

---

10. Reset, power-good, watchdog, errors

10.1 Resets

Event	How	Effect
FPGA configuration	Reset Release IP holds `ninit_done`=1 → `sys_reset_n`=0	Nios, glue and Caliptra requests all held
Board reset	`reset_n`=0	Same as above; also a Caliptra **cold** reset (pwrgood/rst_b requests gated by `sys_reset_n`)
Caliptra cold reset	Host clears `ctrl[0]` (and `[1]`), then sets `ctrl=1` → wait → `ctrl=3`	pwrgood-domain state cleared (fuses, owner hash + lock, AxUSER locks) → full fuse push needed
Caliptra warm reset	Host clears only `ctrl[1]`, then sets it again	Fuses/owner hash persist; host must issue fuse ctrl **START_DONE** in `BOOT_FUSE`. **Not simulated in 2.x**

- Both Caliptra resets use `reset_sync.sv` cells: asynchronous assertion, 2-stage synchronous deassertion, `SYNCHRONIZER_IDENTIFICATION FORCED_IF_ASYNCHRONOUS`. This meets IS Table 4 / the RDC rules.
- In the sim, pwrgood is raised and then `rst_b` follows 20 cycles later (`cold_release()`). That is above IS Table 19's 10-clock minimum.

10.2 Watchdog

- The host writes `CPTRA_WDT_CFG[0] = 100,000,000` and `CPTRA_WDT_CFG[1] = 0` before fuses.
- The debug-locked ROM uses them: the log shows `[state] Starting the WD Timer 100000000 cycles` at cycles 384,577 and 450,858.
- There are no WDT pins. A cascaded timeout shows up as `cptra_error_fatal`.

10.3 Errors

- `cptra_error_fatal` / `_non_fatal` are polled on `status_pio[5:6]`.
- The TB host clears non-fatal errors by RW1C (`HW/FW_ERROR_NON_FATAL`), and treats any fatal as a test FAIL. It dumps BOOT/FLOW status and the error registers.
- **The spec rule is to recover from a fatal error only by a pwrgood cycle.** The 1.x firmware does this (≤3 retries). The 2.x TB does **not** exercise fatal recovery.
- At `ready_for_runtime`, every HW/FW fatal and non-fatal register was 0, and `CPTRA_FLOW_STATUS = b8000000`.

---

11. Boot sequence



Host sequence, step by step. This is the TB behavioral host `nios_caliptra2x_sys`, which is the reference for the future `main.c` port.

1. Wait until the Reset Release IP releases `sys_reset_n`.
2. `ctrl = 1` (pwrgood), wait 20 clk, `ctrl = 3` (rst_b).
3. Poll `status[0]` (`ready_for_fuses`). Read `HW_REV_ID` (expect `0x212`), `HW_CONFIG` (expect `iTRNG_en=0`, `SUBSYSTEM_MODE_en=0`) and `CPTRA_SECURITY_STATE` (expect `7`).
4. Program and lock the AxUSERs: `FUSE_VALID_AXI_USER=2` + lock, `TRNG_VALID_AXI_USER=1` + lock, `MBOX_VALID_AXI_USER[0]=1` + lock. Then `TIMER_CONFIG=20000` and `WDT_CFG0=100,000,000`, `WDT_CFG1=0`.
5. Fuse ctrl: check ID `0x46555345`, write `CTRL=1` (START), poll `STATUS` until !BUSY and (DONE|ERR). Expect `COUNT=118`, then confirm `FUSE_WR_DONE=1` and `OWNER_PK_HASH_LOCK=1`.
6. Throughout steps 6–9: service `etrng_req` (§9), RW1C-clear non-fatal errors, and stop on fatal.
7. Poll `status[1]` (`ready_for_mb_processing`). Read `FW_BYTES` from fuse ctrl `0x10`.
8. Poll `MBOX_LOCK` until bit 0 = 0, and check `MBOX_USER = 1`. Write `MBOX_CMD=0x46574C44` and `MBOX_DLEN=253440`. Then for each of 63,360 words: read `soc_fw_store[i]` and write `MBOX_DATAIN`. Then write `MBOX_EXECUTE=1`.
9. Poll `MBOX_STATUS[3:0]` until non-zero, expect 2 (CMD_COMPLETE), then write `MBOX_EXECUTE=0`.
10. Poll `status[2]` (`ready_for_runtime`), then write magic `0xCA11F00D` (on the LEDs: `0x0D`).

The magic/LED codes the TB uses are 1 = start, 2 = ready_for_fuses, 3 = AxUSER programmed, 4 = fuses done, 5 = ready_for_mb_processing, 6 = FW_LOAD issued, 7 = mailbox complete, `0xCA11F00D` = runtime.

Note on clocks. The TB clock is 100 MHz (`#5`), but `TIMER_CONFIG=20000` ps assumes 50 MHz. This only affects Caliptra's MTIME scaling, not the boot result. Set `TIMER_CONFIG` to match the real board clock.

---

12. Changes vs. the 1.x design

Area	1.x (rev 3)	2.x (this design)
Bus	APB, 18-bit, PAUSER	AXI4, 19-bit, AxUSER (32-bit)
Bridge	`avmm_to_apb`	`avmm_to_axi`
Window	256 KiB (18-bit)	1 MiB (20-bit)
Requester IDs	One shared PAUSER = 1	Nios = 1, fuse ctrl = 2 (unique)
Fuses	83 words, data-only image	104 words + 12 owner hash + lock, address/data image, 117 entries

Area	1.x (rev 3)	2.x (this design)
Owner PK hash	Fuse `FUSE_OWNER_PK_HASH`	Register `CPTRA_OWNER_PK_HASH[12]` + `_LOCK` (pwrgood)
Lifecycle	`FUSE_LIFE_CYCLE` + straps	Straps only
Ready signal	`ready_for_fw_push`	`ready_for_mb_processing` (same PIO bit 1)
Mailbox	128 KiB	256 KiB
ROM / ICCM / DCCM	48 / 128 / 128 KiB	96 / 256 / 256 KiB
PQC	LMS optional	ML-DSA-87 (Adams-Bridge) or LMS, selected by `FUSE_PQC_KEY_TYPE`
New tie-offs	—	`strap_ss_*`, `ss_debug_intent`, `ss_ocp_lock_en`, obf UDS/FE, recovery, DMA AXI manager, `cptra_csr_hmac_key` (new strap)
FW bundle	109,000 B	253,440 B

13. Compliance checklist vs. caliptra-rtl v2.1.2 Integration Specification

References: IS Table 4 (clocks/resets), Table 5 (AXI), Tables 11–13 (straps, generic wires), §TRNG, Table 19 (SoC integration requirements), and `soc_ifc` RDL. The status reflects the **2.x RTL glue plus the TB host sequence**. The Nios firmware for 2.x is not ported yet.

Requirement	Spec ref	Implementation	Status
Clock running before / ≥10 clk around `cptra_pwrgood`	Table 19 Resets and Clocks	Free-running clk; TB 20 clk pwrgood→rst_b	**PASS** (sim)
Async assert / sync deassert of pwrgood, rst_b	Table 4 / RDC	`reset_sync.sv` 2FF cells	**PASS**
Agilex Reset Release before user logic	Intel A5E	`rst_release` gates `sys_reset_n`	**PASS** (stub in sim)
AXI single outstanding, in order, no narrow `mbox_datain`/`TRNG_DATA`	IS Unsupported features	`avmm_to_axi` single-beat, `WSTRB=F`	**PASS**
Drive AxUSER on all requests; never `0xFFFFFFFF`	Table 19 AXI USER	Nios 1, fuse 2	**PASS**
AxUSER unique per agent, non-zero	Table 19 AXI USER	1 vs 2	**PASS**
Mailbox AxUSER programmed + locked before use	Table 19 MAILBOX rules	Step 4, before any mailbox access	**PASS** (sim; ideally done by immutable SoC ROM)
TRNG AxUSER programmed + locked	Table 19 TRNG rules	Step 4	**PASS**
Fuse AxUSER locked or hardened	Table 19 FUSE rules	Programmed + locked to 2; `CPTRA_SET_FUSE_AXI_USER_INTEG=0`	**PASS** (programmable; hardening optional)
Fuse writes + `FUSE_WR_DONE` by immutable logic or ROM	Table 19 Fuses	`soc_fuse_ctrl2x` FSM, but the image is a rebuildable hex in FPGA RAM	**TEST-ONLY**
UDS/FE from OTP, no debug inject path	Table 19 Fuses	STUDY values in a hex image	**TEST-ONLY**
Debug read paths for fuses disabled in Production	Table 19 Fuses	Fuse range gated to user 2; UDS/FE write-only; but the hex image is readable in the bitstream	**TEST-ONLY**
Owner PK hash written + locked before mutable code	`soc_ifc_external_reg.rdl`	Fuse ctrl writes + locks (pwrgood-sticky); Nios could write it in the window before START	**PASS** (study); production: write from immutable logic before the host CPU can reach AXI
Warm reset: SoC re-writes `FUSE_WR_DONE` only	IS Fuses / Boot FSM	`START_DONE` implemented	**OPEN** (not simulated in 2.x)
Security state from the SoC's security state	Table 19 Security State	Hardwired Production + debug locked	**TEST-ONLY**
Obfuscation key: NIST-generated, not software, off scan, stable before reset	Table 19 Obfuscation Key	Fixed study localparam	**TEST-ONLY**
CSR HMAC key generated/rotated per project	Table 19 CSR HMAC Key	Fixed study constant (latched only in MANUFACTURING)	**TEST-ONLY**
Subsystem straps / `ss_*` / obf UDS/FE / recovery tied 0 in passive	Tables 11–12	Tied in `caliptra_passive_top`	**PASS**
`ss_ocp_lock_en` constant	Table 11	`1'b0`	**PASS**
DMA AXI manager not connected in passive	IS DMA / AXI arbitration	Ready/valid tied 0	**PASS**
`generic_input_wires` bit 0 = zeroization status; others 0	Table 13	`64'b0`	**PASS**
TRNG serviced by a hardware gasket (FIPS)	IS §TRNG	Host firmware/TB model, pseudo-random data	**TEST-ONLY**

Requirement	Spec ref	Implementation	Status
Report Caliptra error outputs	Table 19 Error reporting	Polled on `status_pio[5:6]`; not a Nios IRQ	**OPEN** (IRQ)
Recover fatal only via pwrgood reset	Table 19 Error reporting	Planned (1.x has it); not in the 2.x TB	**OPEN**
SoC programs WDT before debug-locked ROM	HW spec WDT	`WDT_CFG` written before fuses; ROM log confirms	**PASS**
`TIMER_CONFIG` programmed	`soc_ifc` RDL	20000 ps (50 MHz)	**PASS** (TB clock is 100 MHz; set per board)
`scan_mode` 0 functional; 1 for 5,000 clk before scan	Table 19 DFT	Tied 0, no scan	**PASS** (FPGA)
Caliptra TAP as a TAP endpoint	Table 19 DFT	Tied off	**OPEN**
Input clock from an on-die oscillator; clock-attack defence	Table 19 Resets and Clocks	Board clock	**OPEN**
No independent power-gating; clock gating via Caliptra FW	Table 19 Power saving	Clock ungated	**PASS**
SRAM zeroization on pwrgood	IS SRAM	Not implemented (bitstream-time zero only)	**OPEN**
Synthesis integrity (`u__size_only__` prims preserved)	IS Synthesis constraints	Not yet addressed in the Quartus flow	**OPEN**
Device fits	—	126 % ALMs on A5E 065	**FAIL** (§14)

Summary: 32 rows: 17 PASS, 7 OPEN, 7 TEST-ONLY, 1 FAIL (fit). Every TEST-ONLY item is intentional for a study and must be replaced for a product.

---

14. A5E 065 fit result and options

From `docs/P1_FIT_RESULT_2X_A5E.md`. Quartus Prime Pro 25.3.0 b109, Analysis & Synthesis only. The top is `rtl_2x/caliptra_passive_top.sv` (Caliptra core only, **no Nios**). `MLDSA_MASKING` is on (upstream default).

Resource	2.x synth	A5EC065	%	1.x (same flow)
ALMs needed (pre-fit estimate)	**279,685**	**222,400**	**125.8 %**	138,715
Combinational ALUTs	395,094	—	—	218,761
Dedicated logic registers	411,569	889,600	46.3 %	141,933
Block memory bits	9,005,696	31.46 Mb	≈ 28.6 %	≈ 5.67 Mb
M20K (estimate)	≈ 484	1,611	≈ 30 %	—
DSP (synth / est. post-merge)	213 / 199	846	25.2 % / 23.5 %	49

- **Biggest consumer: Adams-Bridge** (`abr_inst`): 101,640 ALUTs, 244,925 FFs, 164 DSP. The masked NTT `hybrid_bf2x2` alone has 218,303 FFs.
- **Next:** `ecc_top1` (79,991 ALUTs), VeeR (55,132), key vault (51,019).
- **Over by 57,285 ALMs.** Even a 20 % packing gain leaves the design at ≥ 100 %. The Nios system would add more.

Options

Option	Fits	Spec-conformant	Comment
A. Agilex 7 (e.g. AGF/AGI 014 ≈ 487k ALMs, ~57 % on this estimate; 022/027 for margin)	Yes	Yes	Same RTL; change FAMILY/DEVICE in `quartus_a5e_2x/gen_qsf.py`; new PD/pins
B. Simulation-only 2.x (keep Caliptra 1.x in A5E hardware)	n/a	Yes	Develop the glue, firmware and tooling in Verilator / sw-emulator now
C. Unmasked / reduced test build (`MLDSA_MASKING` undefined and/or `RV_FPGA_OPTIMIZE`)	Unknown, untested	**No**: removes side-channel countermeasures	Per the plan, upstream `MLDSA_MASKING` mainly zeroes the randomness; real savings may need an RTL fork. Research only

---

15. Open items

1. Create the Platform Designer system `nios_caliptra2x_sys`: Nios V/m, on-chip RAM, JTAG UART, ctrl/status/magic PIOs, and an exported 20-bit Avalon `mm_bridge` named `caliptra_axi`.
2. Port `main.c` to the §11 sequence: 2.x register names, AxUSER programming, `ready_for_mb_processing`, 253,440-byte load, warm-reset START_DONE with bridge quiesce, and fatal pwrgood recovery.
3. Add 2.x `warm` and `fatal` Verilator tests, as in the 1.x suite.
4. Choose the hardware target (Agilex 7 vs. sim-only), then do a full fit, timing, and synthesis-integrity constraints.

5. Production replacements for every TEST-ONLY row: OTP fuses, a real lifecycle, an HSM-generated obfuscation and CSR HMAC key, a hardware TRNG, and own signing keys (FIRMWARE_AND_KEYS_2X.md).

Caliptra 2.x + Nios glue — Simulation Guide

> **Where the files are (published copy).** This guide was written in the local Caliptra 2.x design tree. In this repo, the 2.x sources it names live under `caliptra2x/`:
>
> - `rtl_2x/`, `sim_2x/`, `sim/rst_release_stub.sv`, `fw_2x/`, `scripts/gen_fuse_image_2x.py` and `quartus_a5e_2x/` → `caliptra2x/…`
> - `docs/P1_FIT_RESULT_2X_A5E.md` and `docs/CALIPTRA_2X_INTEGRATION_PLAN.md` → this folder
> - The 1.x guide `docs/NIOS_CALIPTRA_INTEGRATION.md` is `../NIOS_CALIPTRA_INTEGRATION.md`
>
> caliptra-rtl and caliptra-sw are not vendored.

- **Scope:** Verilator simulation of the **real** 2.x glue RTL (`rtl_2x/`) around caliptra-rtl v2.1.2 (`49370266`) core-only. It runs the real ROM `rom-2.1.3-1` and the real FW bundle `fw-2.1.4` (FMC + runtime), and boots from cold power-on to `ready_for_runtime`.
- **Result: PASS**: `TB_NIOS2X PASS cyc=36617987 trng=217 (prod, debug locked)`.
- **Related:** NIOS_CALIPTRA_2X_INTEGRATION.md (design), FIRMWARE_AND_KEYS_2X.md (ROM, bundle, fuse image). The raw record is in `sim_2x/BOOT_NIOS2X.md` and `sim_2x/nios2x_top_fw_run.log`.

---

1. What is simulated

Part	Real RTL or model
`nios_caliptra2x_top` (window decode, `soc_fw_store`, `soc_fuse_ctrl2x`, `avmm_to_axi`, AXI write mux, 2FF reset syncs, straps)	**Real** (`rtl_2x/`)
`caliptra_passive_top` + `caliptra_top` v2.1.2 (VeeR, soc_ifc, crypto, Adams-Bridge, memories)	**Real** (`/workspace/caliptra-rtl`, `rtl_2x/caliptra_passive_top.sv`)
Caliptra ROM / FMC / runtime	**Real** binaries: `fw_2x/caliptra-rom.hex`, `fw_2x/caliptra-fw-bundle.hex`
Fuse image	**Real** generator output: `fw_2x/fuse_image_2x.hex`
Agilex Reset Release IP	Stub `sim/rst_release_stub.sv` (`ninit_done` = 1 for 25 ns)
Platform Designer system `nios_caliptra2x_sys` (Nios V + PIOs + window)	**Behavioral model** in `tb_nios2x_top_fw.sv`, with the same port names. It issues the Avalon accesses the 2.x Nios firmware is meant to issue

The Nios V CPU, Platform Designer interconnect and C firmware are **not** in this sim. Compare the 1.x sim, which also used a behavioral host.

---

2. Tools and inputs

Item	Value
Simulator	**Verilator 5.054** (2026-11-01, conda-forge build), from `/workspace/mamba/envs/eda/bin` (override with `VERILATOR_BIN`)
Build flags (`run_nios2x_top_fw.sh`)	`--binary --threads ${THREADS:-4} -j 8 -O3 -Wno-fatal -Wno-lint -Wno-style -Wno-UNOPTFLAT -Wno-MULTIDRIVEN -Wno-TIMESCALEMOD --top-module tb_nios2x_top_fw -Mdir obj_nios2x -f boot2x.f -o Vtb_nios2x`
Caliptra RTL	`/workspace/caliptra-rtl` at v2.1.2 `49370266d12cb0c4a8f71b3a0ff7e54ba7d4866e`, with submodule `adams-bridge`. **Not in git**; clone it with `git clone --recursive --branch v2.1.2 https://github.com/chipsalliance/caliptra-rtl.git`
File list	`sim_2x/boot2x.f` (641 lines): `+incdir+` paths and 588 caliptra-rtl source files (absolute `/workspace/caliptra-rtl/...` paths), plus the repo files listed below
Repo sources in the file list	`sim_2x/patches/decompress_top.sv`, `rtl_2x/caliptra_veer_sram_export.sv`, `rtl_2x/caliptra_passive_top.sv`, `sim/rst_release_stub.sv`, `rtl_2x/reset_sync.sv`, `rtl_2x/avmm_to_axi.sv`, `rtl_2x/soc_fuse_ctrl2x.sv`, `rtl_2x/soc_fw_store.sv`, `rtl_2x/nios_caliptra2x_top.sv`, `sim_2x/tb_nios2x_top_fw.sv`
Verilator patch	`sim_2x/patches/decompress_top.sv` replaces the adams-bridge original. It changes one line: `` `ifdef VERILATOR `` becomes `` `ifdef CLP_OLD_VERILATOR_WORKAROUND `` around the `abr_piso_multi` rate arrays, so Verilator 5.054 uses the normal 4-entry `INPUT_RATES`/`OUTPUT_RATES` instead of the old 5-entry workaround
Data files (via RTL parameter defaults, relative to `sim_2x/`)	`../fw_2x/caliptra-rom.hex`, `../fw_2x/caliptra-fw-bundle.hex`, `../fw_2x/fuse_image_2x.hex`

In the published copy (`caliptra2x/sim_2x/boot2x.f`), the repo files are already relative (`../rtl_2x/…`, which resolve because the run script `cds` into `sim_2x/`); only the caliptra-rtl paths are absolute. To use a different checkout location, rewrite the absolute paths in `boot2x.f`, for

example: `sed -i 's#/workspace/caliptra-rtl#/path/to/caliptra-rtl#g; s#/workspace/exports/caliptra-a5e-2x-quartus-repo#/path/to/repo#g' boot2x.f`.

---

3. Build and run

Run long sims detached. A full boot takes about 1.4 h of wall time, and a foreground shell or agent session can be interrupted.

```
cd sim_2x
# build (~6 min) + run, detached; output in nios2x_top_fw_run.log, wrapper output in run.out
nohup setsid ./run_nios2x_top_fw.sh +test=all > run.out 2>&1 &

# re-run without rebuilding
BUILD=0 nohup setsid ./run_nios2x_top_fw.sh +test=all > run.out 2>&1 &

# watch progress
tail -f nios2x_top_fw_run.log          # LOG lines, TRNG services, heartbeats, milestones
grep -E 'MILESTONE|PASS|FAIL|TIMEOUT' nios2x_top_fw_run.log
```

Env var	Default	Meaning
`BUILD`	`1`	`0` skips the Verilator build
`THREADS`	`4`	Verilator `--threads`
`VERILATOR_BIN`	`/workspace/mamba/envs/eda/bin`	Prepended to `PATH`

- The script runs the binary under `stdbuf -oL -eL`, so the log is line-buffered and `tail -f` works. When the run ends, it appends `wall_time_s=<s> end <date>`.
- **Measured costs** on this box:
  - Build: Verilator walltime **358 s** (elab 4.9 s, convert 204.8 s, C++ build 143.9 s); 2,185 MB allocated; 134 C++ files.
  - Run: **5,069 s wall (1 h 24 m 29 s)** in the wrapper. Verilator reported walltime 4,452.7 s, cpu 17,793.6 s on 4 threads, 18 MB allocated, about 82 µs of sim time per wall second.

3.1 Plusargs

Plusarg	Default	Effect
`+test=axiuser`	—	Cold release, `ready_for_fuses`, AxUSER programming, then the AxUSER checks (§4.1), then `$finish`. Takes a few hundred cycles
`+test=prod`	—	Cold boot to `ready_for_runtime` **without** the AxUSER checks
`+test=all`	**default**	AxUSER checks, then the prod cold boot. **Not** a multi-boot suite: there is no warm or fatal second boot, unlike 1.x `+test=all`
`+max_cycles=N`	300,000,000	Prints `TB_NIOS2X TIMEOUT magic=… trng=…` and stops
`+hb=N`	1,000,000	Heartbeat line period: `[cyc] heartbeat status=<pio[7:0]> trng=<n>`

The TB clock is 100 MHz (`always #5`), so 1 cycle = 10 ns of sim time. `$finish` at 366 ms equals 36.6 M cycles.

---

4. What the testbench checks

4.1 AxUSER test (`test_axiuser`, run by `axiuser` and `all`)

This runs after the host programs and locks the AxUSERs (fuse = 2, TRNG = 1, mailbox[0] = 1). The TB `forces dut.nios_axi_user` and reads the bridge's `err_q` (SLVERR).

Access	AxUSER	Expected	Result (log)
Write `MBOX_LOCK`	3	SLVERR	`[115] MBOX write AxUSER=3 -> SLVERR OK`
Write `FUSE_UDS_SEED[0]`	3	SLVERR	`[121] fuse write AxUSER=3 -> SLVERR OK`
Write `FUSE_UDS_SEED[0]`	1 (Nios)	SLVERR (only user 2 may touch fuses)	`[127] … SLVERR OK`
Read `CPTRA_HW_REV_ID`	1	Accepted	`[132] AxUSER=1 HW_REV_ID=00000212 accepted OK` → `TB_AXIUSER PASS`

4.2 Boot checks (`prod`, `all`)

Check	FAIL message if it fails
`generic_input_wires === 0`	`TB_NIOS2X FAIL generic_input_wires`
Fuse ctrl ID = `0x46555345`	`FAIL fuse id`

Check	FAIL message if it fails
Fuse ctrl ends DONE, not ERR	`FAIL fuse ctrl ERR (SLVERR on a fuse write?)`
Fuse ctrl `COUNT == 118` (104 fuse + 13 owner hash/lock + WR_DONE)	`FAIL expected COUNT=118 …`
`CPTRA_FUSE_WR_DONE[0] == 1`	`FAIL FUSE_WR_DONE not set`
`CPTRA_OWNER_PK_HASH_LOCK[0] == 1`	`FAIL owner PK hash not locked`
`MBOX_USER == 1` after the lock is acquired	`FAIL MBOX_USER`
`MBOX_STATUS == 2` (CMD_COMPLETE) after FIRMWARE_LOAD	`FAIL mbox status` (plus a register dump)
No `cptra_error_fatal` at any time	`FAIL unexpected cptra_error_fatal` (plus a dump)
`etrng_req` drops within 100,000 cycles after each `DATA_WR_DONE`	`FAIL etrng_req stuck …`
`ready_for_runtime` before `max_cycles`	`TIMEOUT`

The TB also **prints, without checking**: `HW_REV_ID`, `HW_CONFIG`, `CPTRA_SECURITY_STATE`, `FUSE_PQC_KEY_TYPE` readback, `CPTRA_OWNER_PK_HASH[0]`, and a final dump of BOOT/FLOW status and all four error registers.

PASS criteria: every check above holds, `ready_for_runtime` is reached, and then 20,000 more cycles run with no fatal error. The TB prints `SUMMARY …` and `TB_NIOS2X PASS cyc=<n> trng=<n> (prod, debug locked)`.

4.3 The `FUSE_PQC_KEY_TYPE readback=00000000` quirk

- The image writes `FUSE_PQC_KEY_TYPE = 1` at `0x3038C` (line 91 of the 117-line image, written by the fuse ctrl with AxUSER 2). The TB reads it back **as the Nios (AxUSER 1)** and gets `00000000`.
- **Cause:** in caliptra-rtl `soc_ifc_arb.sv`, `soc_reg_req` for addresses in `SOC_IFC_FUSE_START_ADDR..SOC_IFC_FUSE_END_ADDR` (`0x30200–0x305FF`) requires `valid_fuse_user`. After `CPTRA_FUSE_AXI_USER_LOCK` = 1, that means AxUSER == 2. The Nios read therefore never reaches the register block, and the read data is 0.
- This is the same gate that gives the SLVERR in §4.1. The TB does not check the bridge error flag on this read, so whether that read also returned SLVERR was not recorded.
- **Proof the fuse was effective:** in Production lifecycle, the ROM compares the manifest PQC type with this fuse and would fail with `IMAGE_VERIFIER_ERR_PQC_KEY_TYPE_MISMATCH`. Instead it logged `[fwproc] Img verified ecc=0 MLDSA=0 svn=0 fuse_svn=0` at cycle 17,010,257.
- By contrast, `CPTRA_OWNER_PK_HASH*` (`0x30140`/`0x30170`) and `CPTRA_FUSE_WR_DONE` (`0x300B0`) are outside the fuse range, so the Nios read them back correctly.
- To read the fuse values back for debugging, issue the reads with AxUSER 2 (for example, add a read path to `soc_fuse_ctrl2x`) or before `CPTRA_FUSE_AXI_USER_LOCK` is set.

4.4 Log channel

- The ROM, FMC and runtime print through `generic_output_wires[7:0]`, with bit 8 toggling per character.
- The TB top assembles lines and prints them as `[cycle] LOG <text>`.
- It records the first `[fmc` / `[alias rt` line as `rom_to_fmc` and the first `[rt]` line as `fmc_to_rt`.

---

5. Measured results (`+test=all`, PID 227528, 2026-11-10)

Environment:

- Start 13:53:45 CST, end 15:18:14 CST.
- `HW_REV_ID = 00000212`.
- `HW_CONFIG = 00000010` (`iTRNG_en=0`, `SUBSYSTEM_MODE_en=0`).
- `CPTRA_SECURITY_STATE = 00000007` (lifecycle 3 = Production, debug_locked = 1).

Cycle	Event (from `nios2x_top_fw_run.log`)
40	`MILESTONE ready_for_fuses`
132	`TB_AXIUSER PASS`
136	`FUSE_CTRL NUM_FUSES=117` (START follows)
614	`FUSE_CTRL STATUS=00000002 COUNT=118`
619	`FUSE_WR_DONE=00000001`; `FUSE_PQC_KEY_TYPE readback=00000000` (§4.3); `OWNER_PK_HASH_LOCK=00000001`; `OWNER_PK_HASH[0]=48afdb07`
634	`MILESTONE fuses done via soc_fuse_ctrl2x (COUNT=118)`
378,838	`Running Caliptra ROM ...`
379,157	First TRNG request serviced

Cycle	Event (from `nios2x_top_fw_run.log`)
384,577	`[state] Starting the WD Timer 100000000 cycles`
442,177 / 445,304	`LifecycleState = Production` / `DebugLocked = Yes`
794,349	`ROM Digest: AB0AEDDBE4F660DC3CDB6F3221DCE2ADA1F6D0593A403B0B12D9EB5CC5C632EB`
5,524,388	`[kat] --` (ROM KATs done)
5,542,288	`MILESTONE ready_for_mb_processing trng=21`
6,049,774	`MILESTONE FIRMWARE_LOAD issued, 253440 bytes from soc_fw_store`
14,288,568	`[fwproc] Recv command 0x46574c44`
17,010,257	`[fwproc] Img verified ecc=0 MLDSA=0 svn=0 fuse_svn=0`
17,085,903	`Load FMC at address 0x40000000 len 30760`
17,304,364	`Load Runtime at address 0x40008000 len 205516`
21,319,080	`MILESTONE MBOX_STATUS=2 (2=CMD_COMPLETE)`
27,820,645	`[exit] Launching FMC @ 0x40000000`
27,875,426	`Running Caliptra FMC ...`
30,035,513	First FMC log line (ROM → FMC)
36,003,133	`Caliptra RT` (RT KATs incl. MLKEM1024 follow)
36,217,560	First RT log line `[rt] OCP LOCK Unsupported` (FMC → RT)
36,596,602	`[rt] Disable entropy source`
36,597,923	`[rt] RT listening for mailbox commands...`
36,597,957	**`MILESTONE ready_for_runtime`**
36,617,987	`TB_NIOS2X PASS cyc=36617987 trng=217 (prod, debug locked)`

Summary line: `SUMMARY ready_for_fuses=40 fuses_done=614 first_trng=379157 ready_for_mb_processing=5542288 fw_load_issued=6049774 mbox_complete=21319080 rom_to_fmc=30035513 fmc_to_rt=36217560 ready_for_runtime=36597957 trng_fills=217 fuse_writes=118`

Final dump: `CPTRA_BOOT_STATUS=00000000`, `CPTRA_FLOW_STATUS=b8000000`, `HW_ERROR_FATAL = FW_ERROR_FATAL = HW_ERROR_NON_FATAL = FW_ERROR_NON_FATAL = 00000000`.

Not covered by this run: warm reset (START_DONE), fatal-error pwrgood recovery, mailbox commands after runtime, the real Nios V / PD interconnect, and the A5E Reset Release IP model.

---

6. Waveforms

The 2.x TB does not dump waveforms as built. `boot2x.f` / `run_nios2x_top_fw.sh` do not pass `--trace`/`--trace-fst` (the build log shows `-DVM_TRACE=0 -DVM_TRACE_FST=0`), and `tb_nios2x_top_fw.sv` has no `$dumpfile`/`$dumpvars`.

To get waves (untested for 2.x; a full 36.6 M-cycle dump of the whole design would be very large and much slower):

1. Add `--trace-fst --trace-structs` (optionally `--trace-depth 3`) to the `verilator` line in `run_nios2x_top_fw.sh`.
2. In `module tb_nios2x_top_fw`, add a windowed dump so only the interesting part is recorded, for example around the fuse push:

```
initial begin
  $dumpfile("nios2x.fst");
  $dumpvars(2, tb_nios2x_top_fw.dut);   // glue level; raise depth for Caliptra internals
  #20us $dumpoff;                         // fuses + start of ROM; move/extend as needed
end
```

3. Rebuild (`BUILD=1`) and run with a smaller `+max_cycles` if you only need the beginning.
4. View with GTKWave: `gtkwave sim_2x/nios2x.fst`. Useful signals:
   - `dut.status_in[7:0]`, `dut.ctrl_out[2:0]`, `dut.cptra_pwrgood`, `dut.cptra_rst_b`
   - `dut.fuse_busy`, `dut.u_fuse_ctrl.st`, `dut.u_fuse_ctrl.idx`
   - `dut.awaddr`, `dut.awuser`, `dut.awvalid`/`awready`, `dut.bresp`, `dut.u_bridge.st`, `dut.u_bridge.err_q`
   - `dut.etrng_req`, `dut.generic_output_wires[8:0]`

The 1.x project's waveform flow (FST + GTKWave screenshots) is in the 1.x repo history; this guide makes no claim that it has been redone for 2.x.

---

7. Troubleshooting

Symptom	Likely cause
`FAIL fuse ctrl ERR` / `COUNT` < 118	A fuse write got SLVERR: fuse AxUSER not programmed/locked to 2 before START, or a bad address in the image
`FAIL etrng_req stuck`	TRNG AxUSER not 1/locked, or `DATA_WR_DONE` (2) not written
ROM fatal around image verification	Fuse image doesn't match the bundle: regenerate with `scripts/gen_fuse_image_2x.py` after every bundle rebuild (FIRMWARE_AND_KEYS_2X.md §5)
Mailbox command rejected / `0x01020009`	Mailbox used with the reserved AxUSER `0xFFFFFFFF` or an unlocked user (see the integration plan)
`TIMEOUT` with no progress	Check heartbeats (`status=` bits) and `trng=` count; the run is long (≈ 36.6 M cycles)

Caliptra 2.x — Firmware, Keys and Fuse-Image Preparation

> **Where the files are (published copy).** This guide was written in the local Caliptra 2.x design tree. In this repo, the 2.x sources it names live under `caliptra2x/`:
>
> - `rtl_2x/`, `sim_2x/`, `sim/rst_release_stub.sv`, `fw_2x/`, `scripts/gen_fuse_image_2x.py` and `quartus_a5e_2x/` → `caliptra2x/…`
> - `docs/P1_FIT_RESULT_2X_A5E.md` and `docs/CALIPTRA_2X_INTEGRATION_PLAN.md` → this folder
> - The 1.x guide `docs/NIOS_CALIPTRA_INTEGRATION.md` is `../NIOS_CALIPTRA_INTEGRATION.md`
>
> caliptra-rtl and caliptra-sw are not vendored.

- **Applies to:** `fw_2x/` and `scripts/gen_fuse_image_2x.py` in this repo.
- **Source:** caliptra-sw `dd1e79d5bf55a9bd5bdb05915757eaf34844145d`, tags `rom-2.1.3-1` and `fw-2.1.4` (checked out on the build box at `/workspace/caliptra-sw-rom`).
- **Hardware revision:** `--hw-revision 2.1`, matching caliptra-rtl v2.1.2 (`CPTRA_HW_REV_ID` reads `0x212`).
- **Related:** NIOS_CALIPTRA_2X_INTEGRATION.md §8 (fuse map), SIMULATION_2X.md.

> ⚠️ **The firmware bundle in `fw_2x/` is signed with caliptra-sw's public "fake" test keys.** The private halves of every vendor and owner key are published in caliptra-sw `image/fake-keys/src/lib.rs`, for example `VENDOR_ECC_KEY_0_PRIVATE`, `VENDOR_MLDSA_KEY_0_PRIVATE`, `OWNER_ECC_KEY_PRIVATE` and `OWNER_MLDSA_KEY_PRIVATE`. A device fused with the hashes in `fw_2x/fuse_image_2x.hex` will accept **any** firmware anyone signs with those keys. Use them for simulation and FPGA study only. §6 describes the production procedure.

---

1. Artifacts in `fw_2x/`

File	Size	What it is
`caliptra-rom.bin`	98,304 B (96 KiB)	2.x ROM **with UART logging** (`ROM_WITH_UART`). SHA-256 `26d505a7ec938d14adbc82627cebd00e3240f7fdd3da464bb4a325c82e18cef2`
`caliptra-rom.hex`	12,288 lines	ROM as little-endian **64-bit** words, one per line (`$readmemh` into `caliptra_rom_m20k`)
`caliptra-fw-bundle.bin`	253,440 B	Signed bundle: manifest + FMC + runtime. SHA-256 `361bdf44d0267dc2f3073f62dc888c94ef929c6095e5d62a615c278b520524f2`
`caliptra-fw-bundle.hex`	63,360 lines	Bundle as little-endian **32-bit** words (`$readmemh` into `soc_fw_store`)
`SHA256SUMS`		`sha256sum *.bin` (covers the two `.bin` files only)
`fuse_image_2x.hex`	117 lines	Fuse image derived from the bundle (§5)
`build_rom.sh`		Rebuild script (§2)

Verify with `cd fw_2x && sha256sum -c SHA256SUMS`. Both hex files were checked word-for-word against the `.bin` files (all 12,288 / 63,360 words match).

---

2. Building the ROM and bundle (`fw_2x/build_rom.sh`)

```
source ~/.cargo/env
SW=${SW:-/workspace/caliptra-sw-rom}          # caliptra-sw at dd1e79d
cd $SW
cargo run --release --manifest-path builder/Cargo.toml --bin image -- --hw-revision 2.1 \
  --rom-with-log $OUT/caliptra-rom.bin --fw $OUT/caliptra-fw-bundle.bin
# then: pad + convert to caliptra-rom.hex (64-bit LE words) and caliptra-fw-bundle.hex (32-bit LE words)
sha256sum *.bin > SHA256SUMS
```

The `image` binary is `builder/bin/image_gen.rs` (`[[bin]] name = "image"` in `builder/Cargo.toml`). With the options above it does the following:

Option	Effect in `image_gen.rs`
`--hw-revision 2.1`	Sets `CALIPTRA_HW_REV=2.1` before building
`--rom-with-log FILE`	`build_firmware_rom(&firmware::ROM_WITH_UART)` (ROM with UART log → the `LOG` lines in the sim)
`--fw FILE`	`build_and_sign_image(&FMC_WITH_UART, &APP_WITH_UART_OCP_LOCK, image_options)`

Option	Effect in `image_gen.rs`
(no `--image-options`)	`ImageOptions { fmc_version: version::get_fmc_version(), app_version: version::get_runtime_version(), fw_svn (default 0), pqc_key_type (default MLDSA), ..Default::default() }`

`ImageOptions::default()` (`builder/src/lib.rs`) supplies the keys:

Field	Default	Meaning
`vendor_config`	`caliptra_image_fake_keys::VENDOR_CONFIG_KEY_0`	4 vendor ECC P-384 keys, 4 ML-DSA-87 keys (32 LMS keys also defined); active `ecc_key_idx = 0`, `pqc_key_idx = 0`; **private keys included**; `pl0_pauser = 0x1`
`owner_config`	`Some(caliptra_image_fake_keys::OWNER_CONFIG)`	1 owner ECC P-384 key + 1 ML-DSA-87 key (+ LMS), **private keys included**
`pqc_key_type`	`FwVerificationPqcKeyType::MLDSA` (= 1; LMS = 3)	
`fw_svn`	0	

Values decoded from the bundle we built (parsed from `caliptra-fw-bundle.bin`):

Field	Value
Manifest marker / size	`0x324E4D43` ("CMN2") / 16,956 B
`pqc_key_type`	1 (ML-DSA)
ECC key descriptor	version 1, `key_hash_count` 4
PQC key descriptor	version 1, key type 1, `key_hash_count` 4
Active vendor key index (ECC / PQC)	0 / 0
Header `flags` / `toc_len` / `pl0_pauser` / `svn`	`0x1` / 2 / `0x1` / 0
FMC TOC	version `0x1044`, load = entry = `0x40000000`, offset 16,956, size 30,760
Runtime TOC	version `0x2010004`, load = entry = `0x40008000`, offset 47,716, size 205,516

The ROM log in the sim agrees: `Load FMC at address 0x40000000 len 30760` and `Load Runtime at address 0x40008000 len 205516`.

---

3. Bundle / manifest layout (2.x, "CMN2")

From caliptra-sw `image/types/src/lib.rs`. Offsets are into `caliptra-fw-bundle.bin`.

Offset	Size	Field
0	4	`marker` = `0x324E4D43`
4	4	`size` = 16,956
8	1 + 3	`pqc_key_type`, reserved
12	**1,736**	`preamble.vendor_pub_key_info` = ECC descriptor (4 + 4×48) + PQC descriptor (4 + 32×48) → **hashed into `FUSE_VENDOR_PK_HASH`**
1,748	4	`vendor_ecc_pub_key_idx`
1,752	96	`vendor_ecc_active_pub_key` (x, y)
1,848	4	`vendor_pqc_pub_key_idx`
1,852	2,592	`vendor_pqc_active_pub_key`
4,444	96 + 4,628	`vendor_sigs` (ECC r|s, PQC signature)
9,168	**2,688**	`owner_pub_keys` = ECC (96) + PQC (2,592) → **hashed into `CPTRA_OWNER_PK_HASH`**
11,856	96 + 4,628	`owner_sigs`
16,580	8	reserved
16,588	160	`header` (revision, key idx, flags, toc_len, pl0_pauser, toc_digest, **svn @ +76**, vendor_data, owner_data @ +120)
16,748	2 × 104	TOC: FMC, runtime
16,956	30,760	FMC image
47,716	205,516	Runtime image
253,232	208	padding to the 253,440-byte bundle size

What gets signed (`image/gen/src/generator.rs`):

- **Vendor:** ECC-P384 signs SHA-384(`header[0..120]`), i.e. the header up to `owner_data` (bundle bytes 16,588–16,707). ML-DSA-87 signs those raw 120 bytes (pure ML-DSA, empty context).
- **Owner:** ECC signs SHA-384(the full 160-byte header, bytes 16,588–16,747). ML-DSA signs those raw bytes.
- The header's `toc_digest` covers the TOC, and each TOC entry's digest covers its image.
- Keys and signatures are stored in "hw format": 32-bit words, little-endian in the file, most-significant word first.

- *Verified:* the vendor and owner ECC signatures in our bundle check out against the active public keys over exactly these byte ranges. The ML-DSA ranges are taken from the generator source and were not independently verified.

---

4. Byte-to-hex conversion

`build_rom.sh` (inline Python):

- **ROM:** pad to a multiple of 8, then write each 8 bytes as `%016x` of a little-endian u64. That gives 12,288 lines for `CALIPTRA_IMEM_DEPTH` = 12288 × 64-bit.
- **Bundle:** pad to a multiple of 4, then write each 4 bytes as `%08x` of a little-endian u32. That gives 63,360 lines. `soc_fw_store` has 65,536 words, and the Nios streams words 0..63,359 into `MBOX_DATAIN`.

---

5. Fuse image generation (`scripts/gen_fuse_image_2x.py`)

```
python3 scripts/gen_fuse_image_2x.py [path/to/caliptra-fw-bundle.bin]   # default fw_2x/caliptra-fw-bundle.bin
# -> fw_2x/fuse_image_2x.hex: 117 entries (104 fuse words + 13 owner-hash/lock); +FUSE_WR_DONE = 118 AXI writes; pqc_key_type=1
```

What the script does:

1. **Checks the bundle.** It asserts:
   - marker = `CMN2`;
   - `pqc_key_type` ∈ {1, 3};
   - `manifest.size` = 16,956 (the 2.x layout);
   - header SVN = 0 (it refuses otherwise, because the SVN fuses are written as 0);
   - the PQC descriptor's key type equals the manifest `pqc_key_type`.
2. **Vendor hash.** `FUSE_VENDOR_PK_HASH` = SHA-384(bundle[12 : 12+1736]), written as 12 **big-endian** 32-bit words. This matches caliptra-sw `image_pk_desc_hash` / `bytes_to_be_words_48`, and the `caliptra-image-app --print-hashes` computation (`image/app/src/create/mod.rs`).
3. **Owner hash.** `CPTRA_OWNER_PK_HASH` = SHA-384(bundle[9168 : 9168+2688]), same word order. Then `CPTRA_OWNER_PK_HASH_LOCK` = 1.
4. **PQC type.** `FUSE_PQC_KEY_TYPE` = manifest `pqc_key_type` (1).
5. **STUDY seeds.** STUDY UDS (16 words) and FE (8 words) from fixed formulas, **not secrets**. Every other fuse word is 0, except `FUSE_SOC_STEPPING_ID` = 1.
6. **Output.** Writes `{addr,data}` lines (see the integration guide §8 for the full table) and prints both hashes. With the current bundle:
   - `vendor_pk_hash 30399676 a17e3e97 3677b3ff 862f4bf2 d1932d88 4778453c 376fe00d c93fb8aa 0770f3eb f3411a08 53e9c57e ce8a2980`
   - `owner_pk_hash 48afdb07 3c5e0d4e e4649046 8ef81f2c f57249b6 e76a28f5 fca4de69 6a7d3e2e d3efc4e6 77431854 3e95307a 54988bd7`

Because the hashes come from the bundle itself, **re-run the script after every bundle rebuild or re-sign**, then rebuild the sim or bitstream (the image is `$readmemh`'d at elaboration). Re-running it on the committed bundle reproduces `fuse_image_2x.hex` byte for byte.

What the ROM checks against these values (`image/verify/src/verifier.rs`, when not Unprovisioned):

- `FUSE_PQC_KEY_TYPE` must equal the manifest PQC type (`IMAGE_VERIFIER_ERR_PQC_KEY_TYPE_MISMATCH`).
- `FUSE_VENDOR_PK_HASH` must be non-zero and match the descriptor digest.
- The active vendor key's hash must appear in the descriptor, and the key must not be revoked.
- The owner key digest must match `CPTRA_OWNER_PK_HASH`.
- Signatures must verify.
- Image SVN must be ≥ fuse SVN (unless anti-rollback is disabled).

---

6. Production key preparation (procedure)

Steps marked *(untested here)* follow the caliptra-sw source but were not run in this project.

6.1 Decide the key set

Key	Algorithm	Count	Held by
Vendor firmware-signing keys	ECC P-384	1–4 (`VENDOR_ECC_MAX_KEY_COUNT` = 4)	Silicon or firmware vendor HSM

Key	Algorithm	Count	Held by
Vendor PQC keys	**ML-DSA-87** (`pqc_key_type` = 1) **or** LMS (`= 3`)	ML-DSA 1–4 (`VENDOR_MLDSA_MAX_KEY_COUNT`), LMS 1–32 (`VENDOR_LMS_MAX_KEY_COUNT`)	Vendor HSM
Owner key	ECC P-384 + the same PQC type	1 each	Platform owner HSM

- All vendor public keys are hashed into one descriptor, so `FUSE_VENDOR_PK_HASH` commits to the whole set.
- Keep spare keys for revocation. The **last** key index can never be revoked (`verify_vendor_ecc_pk_idx`: "Last key index is never revoked").

6.2 Generate keys in an HSM

- Generate every private key **inside the HSM** (non-exportable). Export only the public keys.
- Formats the caliptra-sw tools read (`image/gen/src/lib.rs`, `image/crypto/src/openssl.rs`):

Key	Format accepted
ECC P-384 public	PEM `PUBLIC KEY` (SPKI)
ECC P-384 private (only for local test signing)	PEM, read via OpenSSL `EcKey::private_key_from_pem`
ML-DSA-87 public	Raw 2,592-byte file, **or** PEM SPKI (the BIT STRING payload is used)
ML-DSA-87 private (local test only)	Raw 4,896-byte expanded key, or PEM PKCS#8 whose privateKey octets are exactly that raw key
LMS public / private	PEM or raw `ImageLmsPublicKey` / `ImageLmsPrivKey` structs

- **Local test keys** with OpenSSL 3.5 (for a dry run of the flow, never for a product):

```
mkdir -p keys && chmod 700 keys            # keys/ is gitignored (see 6.6)
openssl ecparam -name secp384r1 -genkey -noout -out keys/vnd_ecc0_priv.pem
openssl ec -in keys/vnd_ecc0_priv.pem -pubout -out keys/vnd_ecc0_pub.pem
openssl genpkey -algorithm ML-DSA-87 -provparam ml-dsa.output_formats=priv-only -out keys/vnd_mldsa0_priv.pem
openssl pkey -in keys/vnd_mldsa0_priv.pem -pubout -out keys/vnd_mldsa0_pub.pem   # SPKI, accepted as-is
```

- **ML-DSA private key format caveat.** OpenSSL's PKCS#8 for ML-DSA wraps the 4,896-byte key in an inner OCTET STRING: the privateKey field is 4,900 bytes. This was checked with OpenSSL 3.5.7 on the box. The caliptra-sw reader takes those octets directly, so it would get 4,900 bytes and fail. Extract the raw key first, e.g. `python3 -c "import base64,sys;d=base64.b64decode(''.join(l for l in open(sys.argv[1]) if not l.startswith('-----')));open(sys.argv[2],'wb').write(d[-4896:])" keys/vnd_mldsa0_priv.pem keys/vnd_mldsa0_priv.bin` *(untested with the image tool)*.

6.3 Sign the firmware — mechanisms that exist in caliptra-sw `dd1e79d`

A. `caliptra-image-app create` with a key-config TOML (`image/app`).

- This signs **ELF** FMC/runtime with key files listed in a TOML. Arguments:
  - `--pqc-key-type 1|3`, `--key-config <toml>`, `--ecc-pk-idx`, `--pqc-pk-idx`
  - `--fmc <elf> --fmc-version N --fmc-rev <hex>`, `--rt <elf> --rt-version N --rt-rev <hex>`
  - `--fw-svn N --out <bundle>`, optional `--own-from-date/--own-to-date/--mfg-from-date/--mfg-to-date`, `--print-hashes`
- Key-config schema (`image/app/src/create/config.rs`); paths are relative to the TOML file:

```
[vendor]
ecc_pub_keys   = ["vnd_ecc0_pub.pem", "vnd_ecc1_pub.pem", "vnd_ecc2_pub.pem", "vnd_ecc3_pub.pem"]
lms_pub_keys   = []
mldsa_pub_keys = ["vnd_mldsa0_pub.pem", "vnd_mldsa1_pub.pem", "vnd_mldsa2_pub.pem", "vnd_mldsa3_pub.pem"]
ecc_priv_keys   = ["vnd_ecc0_priv.pem", ...]     # optional
mldsa_priv_keys = ["vnd_mldsa0_priv.bin", ...]   # optional
[owner]
ecc_pub_key   = "own_ecc_pub.pem"
lms_pub_key   = ""                               # field is required by the schema; read only for LMS
mldsa_pub_key = "own_mldsa_pub.pem"
ecc_priv_key   = "own_ecc_priv.pem"              # optional
mldsa_priv_key = "own_mldsa_priv.bin"            # optional
```

- If the private keys are omitted, the image is generated with **zero signatures**. The tool needs the private keys locally to sign, so with non-exportable HSM keys use **B** for the signatures.
- Get the ELFs with `image --all_elfs <dir>` (`builder/bin/image_gen.rs`).
- `--fmc-rev`/`--rt-rev`/`--pqc-pk-idx` are declared optional but are unwrapped in `run_cmd`, so pass them.

B. `image` builder with `--image-options`, `--hashes`, and signature override (`builder/bin/image_gen.rs`). This is the external/HSM path.

- `--image-options <toml>` replaces the `ImageOptions` struct (versions, `fw_svn`, `pqc_key_type`, full `vendor_config` / `owner_config` with inline public keys and optional private keys). The reference is `builder/test_data/default_image_options.toml`, which is a copy of the fake-key defaults, *including private keys*. For production, put your public keys in and **leave out** `priv_keys`.

- `--hashes <json>` writes `{"vendor": SHA-384(header[..owner_data]), "owner": SHA-384(header)}`. These are the ECC message digests "for external signing tools".
- `--vendor-sig-override <file>` / `--owner-sig-override <file>` replace `preamble.vendor_sigs` / `owner_sigs` with the bytes of `<file>`, parsed as `ImageSignatures`. That is the ECC signature (r‖s, 96 B, hw format) followed by the PQC signature field (4,628 B), 4,724 B total. The CLI help text still says "ECC … concatenated with an LMS signature"; the struct holds whichever PQC type is selected.

HSM signing procedure (B), (untested here):

1. Build once with your public-key-only `--image-options` plus `--hashes h.json`. The bundle comes out with zero signatures.
2. Sign with the HSM:
   - **ECC:** sign `h.json.vendor` with the active vendor ECC key and `h.json.owner` with the owner ECC key (ECDSA P-384 over the given SHA-384 digest).
   - **ML-DSA-87:** sign the **raw header bytes**, not the digest. Vendor = bundle[16588:16708], owner = bundle[16588:16748] (§3). Pure ML-DSA, empty context. `--hashes` does not export these bytes, so cut them from the step-1 bundle.
3. Assemble the two 4,724-byte `ImageSignatures` files. Convert r, s and the ML-DSA signature into caliptra's word order (§3); the ML-DSA signature is 4,627 B plus 1 pad byte.
4. Rebuild with the same `--image-options` plus `--vendor-sig-override vsig.bin --owner-sig-override osig.bin`. Alternatively, patch the bytes at offsets 4,444 and 11,856 of the step-1 bundle.
   - Rebuilding is only valid if the build is reproducible. caliptra-sw has a consistency test for ELF builds, but this was not checked here.
5. Verify before fusing:
   - Check the ECC signatures with your public keys over the §3 byte ranges (the same check we ran on the fake-key bundle).
   - Boot in Verilator (SIMULATION_2X.md), or in caliptra-sw's emulator, with the regenerated fuse image.

No other external-signing tool exists in caliptra-sw `dd1e79d`: no PKCS#11 / HSM plug-in.

6.4 Regenerate the fuse image

- `python3 scripts/gen_fuse_image_2x.py path/to/your-bundle.bin`. This recomputes `FUSE_VENDOR_PK_HASH` and `CPTRA_OWNER_PK_HASH` from your keys and sets `FUSE_PQC_KEY_TYPE`.
- Cross-check against `caliptra-image-app … --print-hashes` (§6.3 A) if you build with method A.
- In a product:
  - `FUSE_VENDOR_PK_HASH`, `FUSE_PQC_KEY_TYPE`, revocations and SVNs are burned into OTP at manufacturing.
  - `CPTRA_OWNER_PK_HASH` (a register in 2.x) is written and locked from immutable SoC logic or ROM on every cold boot, from owner-controlled storage.
  - The UDS seed and field entropy are generated per device and protected by the obfuscation key. They are **never** the STUDY formulas in the script.
- The script currently asserts bundle SVN = 0. Change its SVN handling (and the fuse values) when you adopt an SVN policy (§6.5).

6.5 SVN and revocation policy

- **FW SVN:**
  - Header `svn` is set by `fw_svn` (`--fw-svn` or the image-options `fw_svn`). Maximum 128 (`MAX_FIRMWARE_SVN`).
  - The fuse SVN is `128 − leading_zeros(FUSE_RUNTIME_SVN[0..3] as u128)`, i.e. the position of the highest burned bit (`drivers/src/fuse_bank.rs`, `first_set_msbit`). Burning bit $n-1$ sets fuse SVN n.
  - The ROM rejects images with `svn` < fuse SVN, unless `FUSE_ANI_ROLLBACK_DISABLE` is set (keep it 0 in production).
  - Bump the SVN only for security fixes, and burn the next bit only after the new image is deployed fleet-wide.
- **Vendor key revocation:**
  - `FUSE_ECC_REVOCATION` (4-bit mask), `FUSE_MLDSA_REVOCATION` or `FUSE_LMS_REVOCATION` (masks); bit *i* revokes vendor key *i*.
  - The last index cannot be revoked.
  - After revoking key *i*, sign with key *i*+1 (`ecc_key_idx` / `pqc_key_idx`).
- **Owner key rotation:** sign with the new owner key, and update the owner hash the SoC writes into `CPTRA_OWNER_PK_HASH`. It is not a fuse in 2.x, so this needs no OTP burn, but the storage must be authenticated.
- `FUSE_FMC_KEY_MANIFEST_SVN`, `FUSE_SOC_MANIFEST_SVN` and `FUSE_SOC_MANIFEST_MAX_SVN` are written 0 in this study.

6.6 Keep keys out of git

- The repo `.gitignore` now ignores `keys/`, `*.key`, `*_priv*.pem`, `*_priv*.bin`, `*.p8`, `*.p12` and `*.pfx`. Even so, never copy private keys into the tree. Keep them in the HSM and keep only public keys and bundles in git.
- The fake-key private halves are already public upstream; they are **not** copied into this repo.
- Review `git status` before every commit when working with real keys.

7. Checklist before you trust a bundle

1. `sha256sum -c fw_2x/SHA256SUMS` passes (or record new sums).
2. The bundle parses with marker `CMN2`, the expected `pqc_key_type`, SVN and TOC sizes (the §2 table, or the asserts in `gen_fuse_image_2x.py`).
3. The fuse image was regenerated from **this** bundle, and both printed hashes match your HSM's public keys.
4. Verilator boot reaches `[fwproc] Img verified …` and `ready_for_runtime` with that fuse image.
5. No private key is in the working tree (`git status`, `git ls-files | grep -i priv`).